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Collaborative Ground-Aerial Multi-Robot System for Disaster Response Missions with a Low-Cost Drone Add-On for Off-the-Shelf Drones

用於災害應變任務之地空協同多機器人系統——搭配適用於市售無人機的低成本無人機外掛模組

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***Abstract*—In disaster-stricken environments, it's vital to assess the damage quickly, analyse the stability of the environment, and allocate resources to the most vulnerable areas where victims might be present. These missions are difficult and dangerous to be conducted directly by humans. Using the complementary capabilities of both the ground and aerial robots, we investigate a collaborative approach of aerial and ground robots to address this problem. With an increased field of view, faster speed, and compact size, the aerial robot explores the area and creates a 3D feature-based map graph of the environment while providing a live video stream to the ground control station. Once the aerial robot finishes the exploration run, the ground control station processes the map and sends it to the ground robot. The ground robot, with its higher operation time, static stability, payload delivery and tele-conference capabilities, can then autonomously navigate to identified high-vulnerability locations. We have conducted experiments using a quadcopter and a hexapod robot in an indoor modelled environment with obstacles and uneven ground. Additionally, we have developed a low-cost drone add-on with value-added capabilities, such as victim detection, that can be attached to an off-the-shelf drone. The system was assessed for cost-effectiveness, energy efficiency, and scalability.**

摘要——在遭受災害的環境中，迅速評估損害、分析環境穩定性，並將資源分配至可能有受困者的最脆弱區域至關重要。這類任務若直接由人員執行，既困難又危險。本文利用地面與空中機器人彼此互補的能力，探討一種空中與地面機器人協同作業的方法來解決此問題。空中機器人具備較大的視野、較快的速度與精巧的體積，負責探索區域並建立環境的三維特徵式地圖圖（map graph），同時將即時影像串流傳送至地面控制站。空中機器人完成探索後，地面控制站處理地圖並傳送給地面機器人。地面機器人具有較長的運作時間、靜態穩定性、酬載投送與遠端通話能力，可自主導航至已辨識出的高脆弱性地點。我們在一個含有障礙物與不平地面的室內模擬環境中，以一架四旋翼無人機與一台六足機器人進行實驗。此外，我們開發了一款低成本的無人機外掛模組，具備受困者偵測等加值功能，可安裝於市售無人機上。本系統已就成本效益、能源效率與可擴展性進行評估。

***Index Terms* — Ground-Aerial Robot Collaboration, Drone Add-on, Disaster Response, Tele-conference, Victim Detection**

關鍵詞——地空機器人協同、無人機外掛模組、災害應變、遠端通話、受困者偵測

I. Introduction

一、引言

Disaster situations typically require immediate responses to search and rescue victims, prevent post-disaster scenarios, and assess damages. However, hazardous conditions such as debris, toxic gases, radioactive materials, etc. can impede human rescuers. Consequently, there is a growing interest in developing robotics technologies to aid in these types of crisis circumstances. In disaster scenarios, multi-robot approaches have advantages over single-robot approaches [1][2][3][4] as robots can perform task distribution, exchange information, and complete tasks more efficiently, resulting in faster completion times and reduced energy consumption.

> 災害情境通常需要立即應變，以搜救受困者、防範災後衍生狀況並評估損害。然而，瓦礫、有毒氣體、放射性物質等危險條件可能阻礙人類救援人員。因此，開發機器人技術以協助處理此類危機情境的興趣日益增加。在災害場景中，多機器人方法相較於單一機器人方法具有優勢 [1][2][3][4]，因為機器人之間可以分配任務、交換資訊，更有效率地完成任務，從而縮短完成時間並降低能源消耗。

Aerial robots are fast and offer a large field of view, but have limited flight time. Ground robots, on the other hand, are optimized for longer missions, can carry larger payloads, and actively interact with the environment, although their field of view can be obstructed. To optimize the use of both robot types, we propose a system[1] where initially, an aerial robot equipped with a custom add-on will explore and map an environment while identifying targets or human victims. During this process, the live visual feed can be observed. Then, the hexapod robot will utilise that map to autonomously reach the identified targets, inspect and interact with them, and deliver medi-packs. Additionally, The hexapod robot is equipped with tele-conferencing equipment to communicate with victims, which is an essential aspect in disaster scenarios.

> 空中機器人速度快且視野寬廣，但飛行時間有限。另一方面，地面機器人針對較長時間的任務最佳化，能攜帶較大的酬載，並能主動與環境互動，不過其視野可能受到遮擋。為了充分發揮這兩類機器人的優勢，我們提出一套系統[1]：首先由配備客製化外掛模組的空中機器人探索並建構環境地圖，同時辨識目標或受困人員；在此過程中可觀看即時影像。接著，六足機器人利用該地圖自主抵達已辨識的目標，對其進行檢查與互動，並投送醫療包。此外，六足機器人配備遠端通話設備，可與受困者溝通，這在災害情境中是不可或缺的一環。

[1] A project video demonstration can be found at: https://www.youtube.com/watch?v=jzN4bU5uXSQ

> [1] 專案示範影片請見：https://www.youtube.com/watch?v=jzN4bU5uXSQ

II. Related Work

二、相關研究

D. Chatziparaschis et al. have presented a ground-aerial collaboration in [2] using a DJI Matrice 100 quadcopter and Aldebaran Nao humanoid robot. The approach involves the humanoid robot planning its own paths using a map provided by the unmanned aerial vehicle (UAV). The UAV creates a 3D map by collecting depth information through a multi-stereo camera system, followed by an octree mapping mechanism. In this approach, the humanoid can detect victims, but the UAV is unable to do so, due to possible occlusions from the multi-stereo camera system's placement structure. Moreover, the humanoid lacks mapping sensors or algorithms, and relies solely on the UAV vision system, which transfers its pose with respect to the UAV with the help of an AR marker mounted on top of the humanoid. Therefore, the approach does not effectively utilize the capabilities of both robots.

> D. Chatziparaschis 等人在文獻 [2] 中以一架 DJI Matrice 100 四旋翼機與一台 Aldebaran Nao 人形機器人展示了地空協同。該方法由人形機器人利用無人機（UAV）提供的地圖自行規劃路徑。UAV 透過多組立體相機系統蒐集深度資訊，再以八元樹（octree）建圖機制建立三維地圖。在此方法中，人形機器人能偵測受困者，但由於多組立體相機系統的安裝結構可能造成遮擋，UAV 無法進行偵測。此外，人形機器人缺乏建圖感測器或演算法，完全仰賴

> UAV 的視覺系統，並藉由裝在人形機器人頂部的 AR 標記，取得其相對於 UAV 的位姿。因此，該方法並未有效發揮兩種機器人的能力。

In [5], an Extended Kalman Filter (EKF) based framework that combines data from a UAV (DJI Matrice 100 equipped with DJI Guidance module with a downward looking sensor unit) and an unmanned ground vehicle (UGV) has been introduced for target searching in complex environments. The UAV first perform a surveying task to identify the target coordinates and then sends them to the UGV. The UGV is a custom-designed platform equipped with four mecanum wheels. UGV moves towards the target while avoiding obstacles after the UAV completes mapping and returns to the initial position. Then, UAV autonomously tracks the UGV using a marker to keep it within the UAV's camera view while following the path towards the goal. The requirement to mark targets with ArUco markers may not be suitable in cluttered environments. The UGV's localization heavily relies on a compass module, which may be affected in areas with magnetic interference.

> 文獻 [5] 提出一套以擴展卡爾曼濾波器（EKF）為基礎的框架，結合 UAV（配備 DJI Guidance 模組與下視感測單元的 DJI Matrice 100）與無人地面載具（UGV）的資料，用於複雜環境中的目標搜尋。UAV 先執行勘測任務以辨識目標座標，再傳送給 UGV。UGV 是一個配備四顆麥克納姆輪的客製化平台。待 UAV 完成建圖並返回起始位置後，UGV 一邊避障一邊朝目標前進。接著，UAV 利用標記自主追蹤 UGV，在其沿路徑前往目標時，將其保持在 UAV 相機視野內。該方法需以 ArUco 標記標示目標，可能不適用於雜亂環境；而 UGV 的定位高度依賴羅盤模組，在有磁場干擾的區域可能受到影響。

An aerial robot which provides a bird's-eye view for the ground robot (KUKA youBot) has been introduced in [6] for search and rescue tasks. Leveraging the bird's eye view, the ground robot will plan its path to goal locations using A^* algorithm, while removing obstacles using its 5-DOF manipulator with a two-finger gripper. The use of AprilTags to identify obstacles and the assumption of obstacles being stationary may limit the practical application of this approach in real world scenarios.

> 文獻 [6] 提出一架為地面機器人（KUKA youBot）提供鳥瞰視角的空中機器人，用於搜救任務。地面機器人利用鳥瞰視角，以 A^* 演算法規劃前往目標位置的路徑，並使用其配備雙指夾爪的 5 自由度機械臂移除障礙物。該方法以 AprilTag 辨識障礙物，並假設障礙物靜止不動，這可能限制其在真實場景中的實際應用。

The system in [7] uses aerial and ground robots to map a multi-story environment collaboratively. A Pelican quadcopter equipped with an Inertial Measurement Unit (IMU) and two tracked robots, Kenaf and Quince were used in the approach. Initially, Kenaf is tele-operated through the multi-story environment while collecting sensory data. Then, the Quince which carries a tele-operated quadcopter and an automated helipad will be released to access locations inaccessible by Kenaf. After Kenaf reaches the locations, the helipad is remotely opened and the quadcopter autonomously takes off and continues the mapping process. Upon completion, the quadcopter is signalled to land autonomously on the helipad. This approach takes a longer time to map an area as the ground robot maps first.

> 文獻 [7] 的系統利用空中與地面機器人協同建構多樓層環境的地圖。該方法使用一架配備慣性測量單元（IMU）的 Pelican 四旋翼機，以及兩台履帶式機器人 Kenaf 與 Quince。首先，以遙控方式操作 Kenaf 穿越多樓層環境並蒐集感測資料；接著釋放載有遙控四旋翼機與自動化停機坪的 Quince，以進入 Kenaf 無法到達的位置。在 Kenaf 抵達這些位置後，遠端開啟停機坪，四旋翼機自主起飛並繼續建圖。完成後，系統發出訊號使四旋翼機自主降落於停機坪。由於先由地面機器人建圖，此方法完成一個區域的建圖需時較長。

The approach in [8] uses a UAV with a downward-facing monocular camera and a steering-based wheeled robot. The UAV measures the obstacles' poses relative to the UGV to build a global map that includes both UGV and obstacles with the help of an EKF. However, UGV must be within UAV's view, limiting the robots' independent navigation.

文獻 [8] 的方法使用一架配備下視單眼相機的 UAV 與一台轉向式輪型機器人。UAV 量測障礙物相對於 UGV 的位姿，並藉由 EKF 建立同時包含 UGV 與障礙物的全域地圖。然而，UGV 必須位於 UAV 的視野內，限制了機器人的獨立導航能力。

Multiple approaches use elevation maps for collaborative localization and navigation. In [9], a drone equipped with a camera and IMU creates an elevation map, used as a reference for the quadrupedal ground robot, StarlETH. The localization method is independent of viewpoints and both robots can create elevation maps with onboard sensors. In [10], a hexacopter collects visual data of an unknown area through a monocular camera, which is used to generate a probabilistic elevation map. Then the ground robot uses it to plan a traversable path by interpreting the elevation map as a traversability map. Moreover, the laser range sensor on the ground robot continuously updates the maps as it moves to plan new paths with updated data.

有多種方法利用高程圖進行協同定位與導航。文獻 [9] 中，一架配備相機與 IMU 的無人機建立高程圖，作為四足地面機器人 StarlETH 的參考。此定位方法與視角無關，且兩台機器人皆能以機載感測器建立高程圖。文獻 [10] 中，一架六旋翼機透過單眼相機蒐集未知區域的視覺資料，用以產生機率式高程圖；地面機器人再將高程圖解讀為可通行性地圖，據以規劃可通行路徑。此外，地面機器人上的雷射測距感測器會在移動過程中持續更新地圖，以便依據更新後的資料規劃新路徑。

While the methodology proposed in [11] for collaborative exploration is promising, it may not be suitable for disaster environments due to highly specific hardware requirements, high costs, and limitations of used ground robots to operate in cluttered environments. It is essential to consider the specific requirements of disaster response and recovery when assessing the suitability of robotic systems.

文獻 [11] 所提出的協同探索方法雖然頗具前景，但由於其硬體需求高度特定、成本高昂，且所使用的地面機器人在雜亂環境中運作受限，可能不適用於災害環境。在評估機器人系統是否適用時，必須考量災害應變與復原的特定需求。

Many of the used aerial robots in these type of collaborative methods are either off-the-shelf drones with specific hardware modifications or custom-built drones. This can make it difficult for users to switch to a different type of aerial robot if needed. To overcome this particular limitation, our method incorporates a cost-effective custom hardware add-on that is equipped with essential sensory devices. This enables a seamless attachment to any off-the-shelf drone that has the capacity to accommodate additional weight. This makes our approach highly scalable and adaptable to a wide range of different aerial robots. We chose a six-legged robot as our ground robot based on an evaluation with other mobile ground robot types. The criteria include speed, traversing capabilities, mechanical complexity, disturbance to the ground, stability on uneven terrains, and adaptability to different terrain conditions.

在這類協同方法中，所使用的空中機器人多半是經過特定硬體改裝的市售無人機，或是客製打造的無人機。這使得使用者在有需要時難以改用其他類型的空中機器人。為克服此一限制，我們的方法採用一個具成本效益、配備必要感測裝置的客製化硬體外掛模組，可無縫安裝於任何具備額外載重能力的市售無人機上，使本方法具有高度可擴展性，並能適用於各式各樣的空中機器人。在與其他類型的移動式地面機器人進行評估比較後，我們選擇六足機器人作為地面機器人。評估準則包括速度、通行能力、機構複雜度、對地面的擾動、在不平地形上的穩定性，以及對不同地形條件的適應能力。

III. Methodology

三、研究方法

This section is divided into four subsections. Section III-A describes the high-level overview of the proposed system, including the used components and communication theories. Section III-B discuss the details of the custom-designed drone

add-on along with the mapping and victim detection aspects. Methodologies behind the six-legged robot and the tele-conferencing feature are described in detail in section III-C. Finally, section III-D provides details in regards to how the collaboration is achieved between the ground and aerial robots.

本節分為四個小節。第 III-A 節說明所提系統的高階概觀，包括所用元件與通訊架構。第 III-B 節討論客製化無人機外掛模組的細節，以及建圖與受困者偵測的相關內容。第 III-C 節詳述六足機器人與遠端通話功能背後的方法。最後，第 III-D 節說明地面與空中機器人之間如何達成協同。

A. System Overview

A. 系統概觀

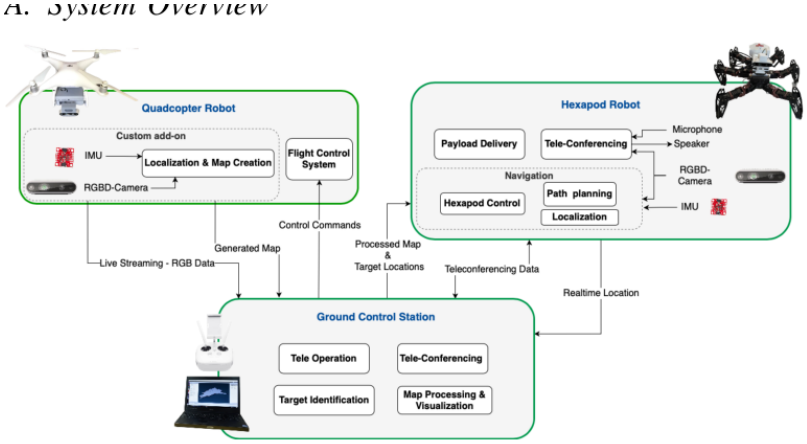


Fig. 1: System Architecture

圖 1：系統架構

The high-level architecture of the collaborative system is comprised of three main components: the quadcopter robot, the hexapod robot, and the ground control station (GCS). A visual representation of the architecture is presented in Figure 1. The quadcopter robot consists of two primary sub-systems: the flight control system and the custom add-on. The flight control system is responsible for controlling the flight of the quadcopter and may include tele-operation, obstacle avoidance, and vision systems depending on the specific quadcopter being used. For our approach, the DJI Phantom 4 Pro drone was selected due to its cost-effectiveness, size, indoor suitability, and precise controllability. The operator at the GCS is able to tele-operate the quadcopter robot using its built-in remote controller. The custom add-on named DroneViz, which is specifically designed with a plug-and-play approach includes mapping and victim detection capabilities and is mounted on the DJI Phantom 4 Pro drone.

此協同系統的高階架構由三個主要元件組成：四旋翼機器人、六足機器人，以及地面控制站（GCS）。架構示意圖如圖 1 所示。四旋翼機器人包含兩個主要子系統：飛行控制系統與客製化外掛模組。飛行控制系統負責控制四旋翼機的飛行，並視所用的四旋翼機型而可能包含遙控操作、避障與視覺系統。在本方法中，我們選用 DJI Phantom 4 Pro 無人機，考量因素為其成本效益、尺寸、適合室內使用以及精準的可控性。GCS 的操作員可使用其內建遙控器遙控操作四旋翼機器人。名為 DroneViz 的客製化外掛模組採隨插即用的設計理念，具備建圖與受困者偵測功能，並安裝於 DJI Phantom 4 Pro 無人機上。

Primary inputs to the DroneViz come from Intel Realsense D435 camera and 9 DoF IMU. These sensors provide visual and inertial data that are used by the localisation and map creation module to perform visual-inertial and feature-based SLAM techniques to create a 3D map graph of the unknown environment. This generated map is transmitted to the GCS for the visualisation and marking of victims. In addition, the live compressed visual feed is streamed over WiFi from the Droneviz to

the GCS for tele-operation and victim detection. This enables the operator to remotely control the quadcopter robot and detect victims in real-time.

> DroneViz 的主要輸入來自 Intel RealSense D435 相機與 9 自由度 IMU。這些感測器提供視覺與慣性資料，供定位與地圖建構模組執行視覺慣性與特徵式 SLAM 技術，以建立未知環境的三維地圖圖。所產生的地圖會傳送至 GCS，用於視覺化與標記受困者。此外，經壓縮的即時影像會透過 WiFi 從 DroneViz 串流至 GCS，以進行遙控操作與受困者偵測，讓操作員能即時遠端控制四旋翼機器人並偵測受困者。

The hexapod Rex is developed using the PhantomX hexapod robot kit by Trossen Robotics and has the same visual-inertial sensory system as DroneViz. The navigation module of Rex obtains the processed map with target locations from the GCS. The map is then used for 3D localisation of Rex and for autonomous navigation. The localisation of Rex is visualised in real-time at the GCS for monitoring. Additionally, the navigation module handles manual control override, navigation stack controls, and other behaviour types of Rex. Dual microphones and stereo speakers are used in a half-duplex (Walkie-talkie) manner for tele-conferencing, and a passive compartment is available for carrying a suitable medipack.

> 六足機器人 Rex 以 Trossen Robotics 的 PhantomX 六足機器人套件開發，並具有與 DroneViz 相同的視覺慣性感測系統。Rex 的導航模組從 GCS 取得已處理並標有目標位置的地圖，再用於 Rex 的三維定位與自主導航。Rex 的定位結果會在 GCS 上即時視覺化以供監控。此外，導航模組也負責處理手動控制接管、導航堆疊控制以及 Rex 的其他行為模式。遠端通話採用雙麥克風與立體聲揚聲器，以半雙工（對講機）方式運作；另有一個被動式置物艙，可攜帶合適的醫療包。

The GCS is the centralised monitoring base station which commands both the quadcopter robot and Rex. The operator can see a live visual feed with victim detection during the operation of the quadcopter robot, and can mark victim positions manually or automatically using the target identification module. After obtaining the map from the DroneViz, the map is processed and visualised as a point cloud for the operator to inspect and identify whether further exploration is needed or to return the quadcopter robot. During Rex's operation, the GCS sends the map to Rex with the target goal locations, and the GCS shows a visual indication of the live location of Rex in the generated 3D map.

> GCS 是集中式監控基地台，負責指揮四旋翼機器人與 Rex。在四旋翼機器人運作期間，操作員可看到帶有受困者偵測結果的即時影像，並可透過目標辨識模組以手動或自動方式標記受困者位置。從 DroneViz 取得地圖後，地圖會經過處理並以點雲形式視覺化，供操作員檢視並判斷是否需要進一步探索，或讓四旋翼機器人返航。在 Rex 運作期間，GCS 將標有目標位置的地圖傳送給 Rex，並在所產生的三維地圖中以視覺方式顯示 Rex 的即時位置。

The system is built upon Robot Operating System (ROS) and the ROS Melodic version is used because of its compatibility with NVIDIA Jetson Nano and high community support. To achieve independence and robustness, a Multi-ROS communication network architecture is implemented. This allows each robot to run its own ROS master and namespace and use ROS topics to communicate between the ROS masters. This approach enables the subsystems to sync only necessary nodes and topics, becoming robust to connection dropouts in disaster scenarios and achieving scalability.

> 本系統建構於機器人作業系統（ROS）之上，並因其與 NVIDIA Jetson Nano 的相容性及龐大的社群支援而採用 ROS Melodic 版本。為達成獨立性與強健性，我們實作了多 ROS（Multi-ROS）通訊網路架構，讓每台機器人執行各自的 ROS master 與命名空間，並透過 ROS topic 在各 ROS master 之間通訊。此方式使各子系統僅需同步必要的節點與 topic，因而能承受災害場景中的連線中斷，並具備可擴展性。

B. DroneViz: Custom Add-on

B. DroneViz：客製化外掛模組

Fig. 2: DroneViz

圖 2：DroneViz（圖中標示：氣流孔、RealSense 相機支架、IMU、SD 卡槽、RealSense 相機、線材槽、融合 LED 狀態指示燈、電源開關、安全關機按鈕、WiFi 天線）

The aerial robot's main duty is to map an area and locate targets while streaming live footage to the control station. DroneViz, depicted in Figure 2, features safety protocols, specialized design aspects, and specific capabilities. The NVIDIA Jetson Nano 4GB board serves as the DroneViz's brain, while WiFi 802.11ac 5GHz is used for communication with the GCS. To supply power to the core device, a 3-cell lithium polymer battery with a 2200 mAh capacity is used, along with a 5V/4A step-down converter module. The core device has a safety shutdown switch and RGB LED indicators for visual cues. The Realsense D435 by Intel and ICM 20948 IMU by Sparkfun Industries serve as DroneViz's primary sensors.

> 空中機器人的主要職責是對區域建圖並定位目標，同時將即時畫面串流至控制站。圖 2 所示的 DroneViz 具備安全機制、專門的設計考量與特定功能。NVIDIA Jetson Nano 4GB 開發板作為 DroneViz 的運算核心，並以 WiFi 802.11ac 5GHz 與 GCS 通訊。核心裝置由一顆容量 2200 mAh 的 3 芯鋰聚合物電池供電，並搭配 5V/4A 降壓轉換模組。核心裝置設有安全關機開關及提供視覺提示的 RGB LED 指示燈。Intel 的 RealSense D435 與 SparkFun 的 ICM 20948 IMU 是 DroneViz 的主要感測器。

The DroneViz model was created utilizing the Solidworks software, where the components were meticulously positioned to ensure that the centre of gravity of the entire DroneViz was not impacted. The placement of components was tested and validated using Solidworks simulations, which ensured that the model was optimized for stability and balance when attached to an aerial robot. The final prototype was 3D printed using acrylonitrile butadiene styrene (ABS). DroneViz costs around US$350 which is much less than the other methods that require building a drone with custom components.

> DroneViz 的模型以 SolidWorks 軟體建立，各元件經過精心配置，以確保不影響整個 DroneViz 的重心。元件配置經 SolidWorks 模擬測試與驗證，確保模型安裝於空中機器人時在穩定性與平衡上達到最佳化。最終原型以丙烯腈-丁二烯-苯乙烯（ABS）3D 列印製成。DroneViz 成本約 350 美元，遠低於其他需以客製元件打造無人機的方法。

In our selected approach, the initial search of the disaster environment is performed by an aerial robot. The aerial robot provides a unique perspective of the environment from a higher altitude, which cannot be replicated by the hexapod robot. Therefore, the use of 2D mapping would be insufficient for the hexapod to localize itself on the map provided by the aerial robot. As a result, a feature-based 3D mapping algorithm called Real-Time Appearance Based Mapping (RTAB-Map)[12] is selected as the preferred approach, allowing the hexapod to localize itself on the map by matching features, even though the viewpoints are different for the two robots.

> 在我們採用的方法中，災害環境的初步搜尋由空中機器人執行。空中機器人從較高處提供獨特的環境視角，這是六足機器人無法複製的。因此，若使用二維建圖，將不足以讓六足機器人在空中機器人提供的地圖上完成自我定位。

基於此，我們選用一種名為即時外觀式建圖（Real-Time Appearance Based Mapping, RTAB-Map）[12] 的特徵式三維建圖演算法，即使兩台機器人的視角不同，六足機器人仍能透過特徵比對在地圖上定位自身。

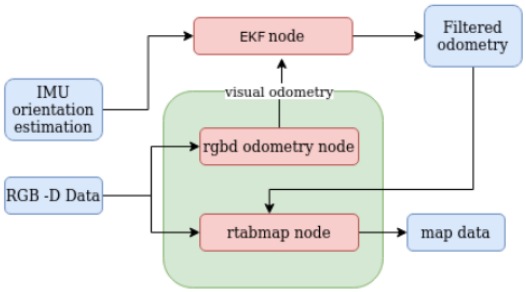


Fig. 3: Diagram for mapping with visual-inertial odometry

圖 3：結合視覺慣性里程計的建圖流程圖（IMU 姿態估測、RGB-D 資料、EKF 節點、濾波後里程計、視覺里程計、rgbd 里程計節點、rtabmap 節點、地圖資料）

The system depicted in Figure 3 utilizes RGB-D data acquired from an Intel Realsense D435 camera to produce visual odometry estimates. These estimates are subsequently used by the mapping node to construct a 3D map of the environment. We achieved 7Hz visual odometry estimates on the Jetson Nano, which met our needs for 3D mapping. However, in order to enhance the reliability and accuracy of our odometry estimates, and to increase the odometry rate, we investigated the use of a visual-inertial odometry approach. This approach fused visual odometry with attitude estimates from an IMU using an EKF. By combining both visual and inertial information, we achieved a more accurate and robust odometry estimate with an increased update rate of 15Hz to roll, pitch, and yaw.

圖 3 所示的系統利用 Intel RealSense D435 相機取得的 RGB-D 資料產生視覺里程計估測值，再由建圖節點據以建構環境的三維地圖。我們在 Jetson Nano 上達到 7Hz 的視覺里程計估測頻率，已能滿足三維建圖的需求。然而，為了提升里程計估測的可靠性與準確度，並提高里程計頻率，我們研究了視覺慣性里程計方法。此方法使用 EKF 融合視覺里程計與 IMU 的姿態估測值。結合視覺與慣性資訊後，我們獲得更準確且強健的里程計估測，翻滾、俯仰與偏航的更新頻率提升至 15Hz。

To obtain an accurate attitude estimation using IMU raw data, we used a quaternion-based complementary filter as it gives more accurate estimates with a faster convergence rate when compared with EKF-based approaches. If only the gyroscope and accelerometer data are provided to the filter, it corrects only the roll and pitch of the predicted attitude estimate. If magnetometer data is also provided, yaw can be corrected as well. Hence, our complimentary filter gives attitude estimation in quaternion form from fusing gyro-scope data with accelerometer and magnetometer data in the form of two delta quaternions in an additional correction step (multiplicative approach) as shown in equation 1.

為了利用 IMU 原始資料取得準確的姿態估測，我們採用以四元數為基礎的互補濾波器，因為與以 EKF 為基礎的方法相比，它能提供更準確的估測且收斂速度更快。若只提供陀螺儀與加速度計資料給濾波器，它只會修正預測姿態中的翻滾與俯仰；若同時提供磁力計資料，則偏航也能被修正。因此，我們的互補濾波器在一個額外的修正步驟中（乘法式方法），將陀螺儀資料與以兩個增量四元數形式表示的加速度計及磁力計資料融合，以四元數形式輸出姿態估測，如式（1）所示。

Here, the initial estimate computed from gyroscope data is depicted as (q_{ω}), and delta quaternions derived from the accelerometer and magnetometer are depicted as Δq_{acc}, Δq_{mag} respectively. This Δq_{mag} is obtained by performing a rotation only about the global z-axis after aligning the global x-axis into the positive direction of the magnetic north pole. Therefore, with this formulation, such rotation does not affect the roll and pitch components even in the presence of magnetic disturbances, limiting the influence only to the yaw angle.

其中，由陀螺儀資料計算的初始估測值記為 (q_ω)，由加速度計與磁力計推導出的增量四元數分別記為 Δq_{acc} 與 Δq_{mag}。Δq_{mag} 是在將全域 x 軸對準磁北極正方向後，僅繞全域 z 軸旋轉而得。因此，在此公式下，即使存在磁場干擾，該旋轉也不會影響翻滾與俯仰分量，影響僅限於偏航角。

$$ {}^{L}_{G}q = {}^{L}_{G}\, q_\omega \Delta q_{acc} \Delta q_{mag} \tag{1} $$

Figure 3 shows the implemented visual-inertial mapping approach for 3D mapping. Visual odometry (VO) output of RTAB-Map and attitude estimation (AE) from IMU data were fused using an EKF to obtain visual-inertial odometry. In this approach, the standard Jacobian matrix of the observation model of an EKF is replaced by a new matrix (H) to allow partial updates of the state vector. Therefore, this allows using sensors that do not measure all the variables in the state vector, which is usually the case. In our case, when measuring only VO and AE, H becomes a 2 by 12 matrix of rank 2 as shown in Table I, with its only non-zero values existing in the columns of the measured variables.

圖 3 呈現了為三維建圖所實作的視覺慣性建圖方法。RTAB-Map 輸出的視覺里程計（VO）與來自 IMU 資料的姿態估測（AE）透過 EKF 融合，以取得視覺慣性里程計。在此方法中，EKF 觀測模型的標準 Jacobian 矩陣被一個新矩陣（H）取代，以允許對狀態向量進行部分更新。因此，即使感測器無法量測狀態向量中的所有變數（這通常是常態），也能加以使用。在本例中，當僅量測 VO 與 AE 時，H 成為一個秩為 2 的 2×12 矩陣（如表 I 所示），其非零值僅出現在被量測變數所對應的欄位。

TABLE I: Matrix H
表 I：矩陣 H

	x	y	z	Φ	θ	ψ	x'	y'	z'	Φ'	θ'	ψ'
VO	1	1	1	1	1	1	1	1	1	1	1	1
AE	0	0	0	1	1	1	0	0	0	0	0	0

Then the generated visual-inertial odometry is fed into the mapping node of the RTAB-Map to generate the 3D map of the environment in real-time. The major benefit of this approach was the ability to improve the odometry rate. The recommended odometry publishing rate for 3D mapping with RTAB-Map is around 15Hz. However, using Jetson Nano, we could only obtain visual odometry at a rate of around 7Hz. As this sensor fusion approach improved the odometry (visual-inertial odometry) rate to the recommended level, we could proceed with indoor mapping faster with fewer distortions. However, the major issue of this approach is that visual odometry works as the primary measurement source of the EKF. Out of all 12 variables of the state vector, only 3 variables (roll, pitch, yaw) are updated using the attitude estimation. Therefore, in case of visual odometry failure, the fusion method is unable to give accurate odometry data which leads to occur distortions in the map. The best way to tackle this issue is to use a tracking camera like Intel RealSense T265 which can directly provide visual-inertial odometry. Nevertheless, we managed this issue by controlling the mapping node in instances where the visual odometry is lost. Simply, whenever the visual odometry is lost during the mapping process, the RTAB-Map node would pause the mapping until the visual odometry gets back on track.

接著，將產生的視覺慣性里程計輸入 RTAB-Map 的建圖節點，即時產生環境的三維地圖。此方法的主要好處在於能提升里程計頻率。以 RTAB-Map 進行三維建圖時，建議的里程計發布頻率約為 15Hz；然而在 Jetson Nano 上，我們只能取得約 7Hz 的視覺里程計。由於此感測器融合方法將里程計（視覺慣性里程計）頻率提升至建議水準，我們得以更快速地進行室內建圖，且地圖失真更少。不過，此方法的主要問題在於視覺里程計是 EKF 的主要量測來源：在狀態向量的全部 12 個變數中，只有 3 個變數（翻滾、俯仰、偏航）由姿態估測更新。因此，一旦視覺里程計失效，融合方法便無法提供準確的里程計資料，導致地圖出現失真。解決此問題的最佳方式是使用如 Intel RealSense

> T265 這類可直接提供視覺慣性里程計的追蹤相機。儘管如此，我們藉由在視覺里程計遺失時控制建圖節點來處理此問題：簡單來說，只要建圖過程中視覺里程計遺失，RTAB-Map 節點便會暫停建圖，直到視覺里程計恢復正常。

1) Victim Detection:

1) 受困者偵測：

Based on the literature survey, seven physical parameters (voice, body heat, scent, motion, skin colour, face/body shape, respiration) were identified to detect victims, and among them, body heat and face/body shape methods were selected for our experiments. Three sensors were chosen to conduct these experiments, namely PIR sensor, infrared camera, and Intel RealSense D435 camera. Out of the three sensors, Intel RealSense D435 was selected due to its wider field of view, high detection range, and high robustness. As to use with the RealSense D435 camera, the YoloV4 model, which supports ROS and provides high FPS for real-time object detection with decent accuracy, was chosen for face/body shape detection.

> 根據文獻調查，我們歸納出七種可用於偵測受困者的生理參數（聲音、體熱、氣味、動作、膚色、臉部／身體外形、呼吸），並從中選擇體熱與臉部／身體外形兩種方法進行實驗。實驗選用三種感測器：PIR 感測器、紅外線相機與 Intel RealSense D435 相機。三者之中，Intel RealSense D435 因具有較寬的視野、較遠的偵測距離與較高的強健性而獲選。為搭配 RealSense D435 相機使用，我們選擇支援 ROS、能以高 FPS 進行即時物件偵測且準確度尚佳的 YOLOv4 模型，用於臉部／身體外形偵測。

2) Pointing Victim Poses in Map:

2) 在地圖上標示受困者位姿：

The victim detection model running at the GCS which leverages the feed from RealSense D435 data will help the controller to identify victims in a disaster area. The controller is capable of manually marking the victim poses using a specific button in the GCS dashboard or can enable automatic marking at the beginning of the mission. The victim marking methodology will use the robot's position and orientation details to mark an arrow pointing to the victim on the map which is detected through the YoloV4 model. This feature is useful for effective navigation and rescue missions.

> 在 GCS 上執行、利用 RealSense D435 影像資料的受困者偵測模型，可協助操控人員辨識災區中的受困者。操控人員可透過 GCS 儀表板上的特定按鈕手動標記受困者位姿，或在任務開始時啟用自動標記。受困者標記方法會利用機器人的位置與朝向資訊，在地圖上標出一個指向由 YOLOv4 模型偵測到之受困者的箭頭。此功能有助於有效導航與執行救援任務。

C. Rex: Hexapod Robot

C. Rex：六足機器人

We have designed a hexapod robot using Phantomx-AX chassis with 18 Dynamixel 12A motors, powered by a 5200 mAh battery. Two Jetson Nano boards have been used for locomotion, localization, navigation tasks, and tele-conferencing with a victim using a speaker and mic. It can carry payloads with a separate enclosure which has been attached to the rear of the robot and provides a live video feed for the operator to observe. Rex weighs 3.51kg and consumes 3.02A when traversing using the tripod gait. Additionally, it can operate for around 1.5 hours with the installed onboard battery.

> 我們以 PhantomX-AX 底盤設計了一台六足機器人，配備 18 顆 Dynamixel 12A 馬達，由一顆 5200 mAh 電池供電。機上使用兩片 Jetson Nano 開發板，負責步態運動、定位、導航任務，以及透過揚聲器與麥克風與受困者進行遠端通話。機器人後方附有獨立置物艙，可攜帶酬載，並提供即時影像供操作員觀看。Rex 重 3.51 公斤，以三足步態行進時耗電 3.02A；使用機上電池約可運作 1.5 小時。

1) Control Aspect of Rex:

1) Rex 的控制：

We developed a kinematic model of Rex using the DH convention and simulated it in the Gazebo physics engine in ROS by leveraging [13]. After validating, we implemented the locomotion engine on Jetson Nano for real-world use. The locomotion engine consists of a trajectory generator, tip pose generator, and a leg controller, customized and fine-tuned for Rex. We implemented and tested four gait patterns: wave, ripple, amble, and tripod. To suit targeted operation environments, we added several additional capabilities such as leg manipulation, and manual body posing on top of the locomotion engine. The Logitech F710 wireless gamepad enables manual control when the Rex reaches the victim location, allowing further inspection of the area. Furthermore, it enables the user to adjust the tip positions of selected legs, which is helpful for clearing objects in the robot's path. Rex's Intel RealSense D435 camera is currently unable to rotate or tilt, which limits its field of view. To address this and enable the robot to traverse in confined spaces, we added manual body posing capability. The robot can control its body along x-y axes, pitch-roll axes, and z axis.

> 我們以 DH 參數法建立 Rex 的運動學模型，並借助 [13] 在 ROS 的 Gazebo 物理引擎中進行模擬。驗證完成後，我們在 Jetson Nano 上實作步態運動引擎以供實際使用。該引擎由軌跡產生器、足端位姿產生器與腿部控制器組成，並針對 Rex 客製化與微調。我們實作並測試了四種步態：波動步態（wave）、漣波步態（ripple）、緩行步態（amble）與三足步態（tripod）。為配合目標作業環境，我們在步態運動引擎之上加入多項功能，例如腿部操作與手動機身姿態調整。當 Rex 抵達受困者位置時，可透過 Logitech F710 無線手把進行手動控制，以進一步檢查該區域；使用者亦可調整選定腿部的足端位置，有助於清除機器人路徑上的物體。Rex 的 Intel RealSense D435 相機目前無法旋轉或傾斜，限制了其視野。為解決此問題並讓機器人能在狹窄空間中通行，我們加入了手動機身姿態調整功能，機器人可沿 x-y 軸、俯仰-翻滾軸以及 z 軸控制機身。

2) Autonomous Navigation:

2) 自主導航：

We implemented a customized navigation stack in ROS that aligns with Rex. The quadcopter's map is fed to the navigation stack, which uses A* algorithm as the global planner and DWA planner as the local planner. Rex can handle dynamic obstacles by updating costmaps with data from the Intel RealSense D435 camera. Localization of Rex in the map provided by DroneViz is performed using feature mapping.

> 我們在 ROS 中實作了一套針對 Rex 客製化的導航堆疊。四旋翼機的地圖輸入導航堆疊，以 A* 演算法作為全域規劃器、DWA 規劃器作為區域規劃器。Rex 可利用 Intel RealSense D435 相機的資料更新代價地圖（costmap），以應對動態障礙物。Rex 在 DroneViz 所提供地圖中的定位，係透過特徵比對完成。

3) Tele-conferencing and Payload Delivery:

3) 遠端通話與酬載投送：

After the evaluation of several methodologies, we have used a Waveshare audio card that can be conveniently installed on the Jetson Nano board. The system was implemented on Rex using the half-duplex walkie-talkie method to reduce communication complexity. An enclosure has been created with a sliding door open mechanism that can be used for payload delivery tasks. The design is a prototype, and the volume can be changed by users to fit their needs while staying under the maximum weight constraint of 200g when using the amble gait.

> 在評估多種方法後，我們採用可方便安裝於 Jetson Nano 開發板上的 Waveshare 音效卡，並以半雙工對講機方式在 Rex 上實作該系統，以降低通訊複雜度。我們另製作了一個具滑門開啟機構的置物艙，可用於酬載投送任務。此設計為原型，使用者可依需求調整容積，但在使用緩行步態時須維持在 200 公克的最大重量限制之下。

D. Ground - Aerial Collaboration

D. 地空協同

Rex can autonomously navigate to target locations by using the 3D map with target locations generated by DroneViz. Before navigation, Rex localizes itself in the map using the localization mode in the RTAB-Map node. Even though the viewpoints of Rex's and DroneViz's cameras differ due to the height difference, RTAB-Map can handle them by using feature mapping.

> Rex 可利用 DroneViz 所產生、標有目標位置的三維地圖，自主導航至目標位置。導航前，Rex 使用 RTAB-Map 節點的定位模式在地圖中定位自身。儘管 Rex 與 DroneViz 的相機因高度差而視角不同，RTAB-Map 仍可透過特徵比對加以處理。

IV. Results

四、實驗結果

In this section, we present the evaluations of the proposed system considering DroneViz and Rex.

> 本節針對 DroneViz 與 Rex，呈現所提系統的評估結果。

Fig. 4: Images taken from the created disaster environment

圖 4：所建置之災害環境的照片

These evaluations were conducted in an indoor environment (Figure 4) that includes walls of different heights, different types of objects, and terrains. This environment was designed to simulate a real-world disaster scenario where the system could be deployed. Additionally, we have performed simulations for Rex in the Gazebo simulator to ensure the algorithms are working as expected before testing in the real world.

> 這些評估在一個室內環境（圖 4）中進行，環境中包含不同高度的牆面、不同類型的物體與地形，旨在模擬本系統可能部署的真實災害場景。此外，在實地測試之前，我們也在 Gazebo 模擬器中對 Rex 進行模擬，以確保演算法運作符合預期。

A. DroneViz Evaluations

A. DroneViz 評估

1) Tests and Parameters:

1) 測試與參數：

The DroneViz's hardware design was validated through various tests, including a maximum load test and a weight test. The maximum load was found to be 2A and the total weight of the DroneViz was 650g. This has a longer operation time of 1 hour compared to the selected drone's (DJI Phantom 4 Pro) flight time of around 30 minutes.

> DroneViz 的硬體設計經由多項測試驗證，包括最大負載測試與重量測試。測得最大負載電流為 2A，DroneViz 總重 650 公克。其運作時間可達 1 小時，長於所選無人機（DJI Phantom 4 Pro）約 30 分鐘的飛行時間。

2) Accuracy of Mapping:

2) 建圖準確度：

Figure 5 shows two 3D maps generated using visual-inertial odometry. Using this approach improved the mapping accuracy when compared with the maps generated using only visual odometry, and the accuracy was further increased with improved visual-inertial odometry by activating internal digital low pass filters of the ICM-20948 IMU which made the output raw data free of high-frequency noise. A ground truth 2D map was created and compared with the 2D map generated by DroneViz as shown in Figure 6.

圖　5　顯示以視覺慣性里程計產生的兩張三維地圖。與僅使用視覺里程計產生的地圖相比，此方法提升了建圖準確度；而在啟用 ICM-20948 IMU 內部的數位低通濾波器、使輸出的原始資料不含高頻雜訊後，改良的視覺慣性里程計進一步提高了準確度。我們建立了一張真值二維地圖，並與 DroneViz 產生的二維地圖比較，如圖 6 所示。

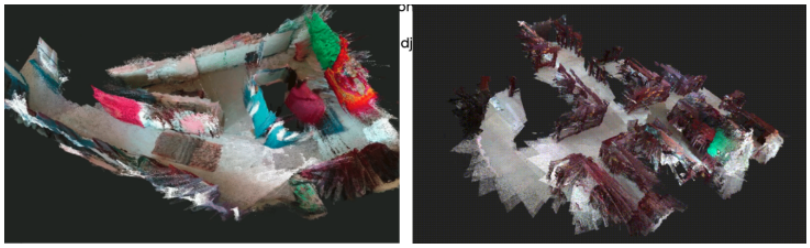


Fig. 5: Improved 3D maps

圖 5：改良後的三維地圖

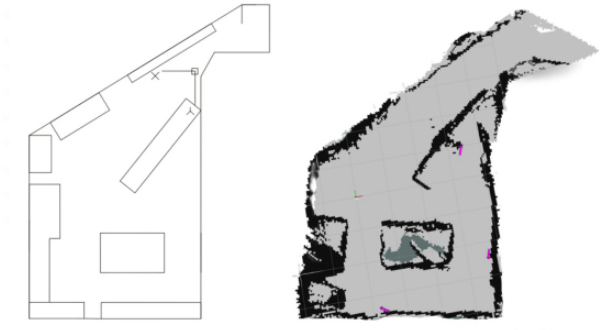


Fig. 6: Comparison of ground truth with 2D test map

圖 6：真值與二維測試地圖之比較

3) Victim Detection Results:

3) 受困者偵測結果：

The performance of the YOLOv4 model was evaluated on the ground control station, using an ASUS TUF F5 laptop. The robustness of the YOLOv4 model is visualized in the figures, and the C++ based Darknet Yolo implementation was used for inference. As shown in Figure 7, the model was able to identify victims even if only a part of the body was within view, and it can identify other objects which make suspicious of whether a victim is nearby.

YOLOv4 模型的效能在地面控制站上以一台 ASUS TUF F5 筆記型電腦進行評估，推論採用以 C++ 實作的 Darknet YOLO。YOLOv4 模型的強健性如圖所示：如圖 7 所示，即使視野中只出現部分身體，模型仍能辨識出受困者，也能辨識其他可能暗示附近有受困者的物體。

Fig. 7: Detecting victims in extreme cases and related objects
圖 7：在極端情況下偵測受困者及相關物體

Marking victims' poses on the map plays an important role in the system. Figure 8 shows some sample maps, marked with victim poses. Arrowheads show the directions of victims from the point where the victim is detected.

在地圖上標記受困者位姿是本系統中的重要功能。圖 8 顯示數張標有受困者位姿的範例地圖，箭頭表示從偵測到受困者的位置指向受困者的方向。

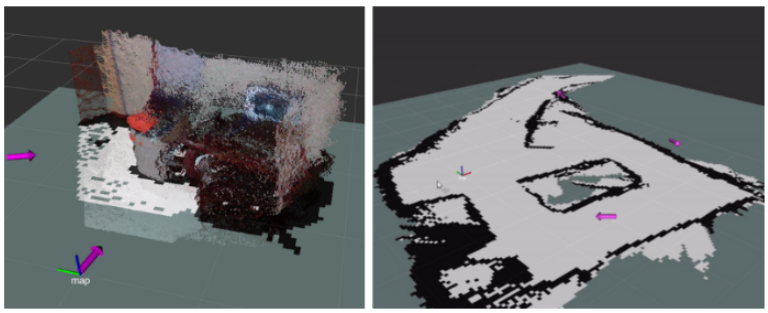


Fig. 8: Marking victims' poses
圖 8：標記受困者位姿

B. Rex Evaluations

B. Rex 評估

Given the intended use of Rex in challenging environments, it is critical to validate its capabilities. To achieve this, a series of tests were conducted and can be broadly classified as current tests, torque tests, and traversability tests.

由於 Rex 預定用於嚴苛環境，驗證其能力至關重要。為此我們進行了一系列測試，大致可分為電流測試、扭矩測試與通行能力測試。

1) Current Tests:

1) 電流測試：

In order to determine the maximum operation time and select a suitable battery, the current consumption of Rex was measured and plotted using DC power supplies. The goal was also to identify body heights that minimize current consumption during stance and gait stages. After conducting several current tests, the lowest current consumption was observed at a body height of 14cm during the stance and 12cm during the gait stage. These parameters were programmed into Rex to minimize current consumption.

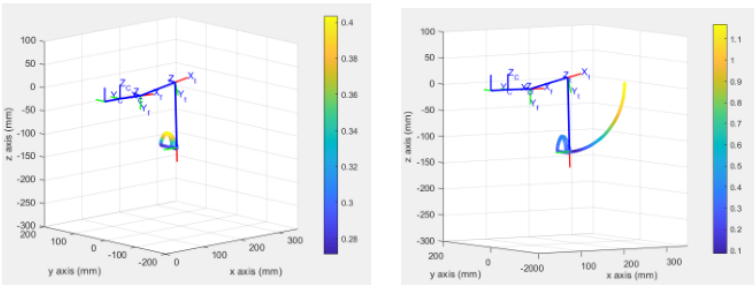
為了確定最長運作時間並選擇合適的電池，我們使用直流電源供應器量測並繪製 Rex 的耗電流，目標也包括找出在站立與步行階段能使耗電流最小化的機身高度。經多次電流測試後發現，站立時機身高度 14 公分、步行時 12 公分的耗電流最低。這些參數已寫入 Rex 的程式中，以將耗電流降至最低。

2) Torque Tests:

2) 扭矩測試：

To ensure the safety and maximum performance of legged robots, it is necessary to determine their maximum payload capacities. The Dynamixel 12A motors used in Rex have a stall torque of 1.5Nm, and calculations were performed for the tripod gait, which is the worst-case scenario when it comes to payload bearing. Analysis as depicted in Figure 9 showed that Rex can safely carry all its components, even in the tripod gait, with a safety margin of 25-33 % from the stall torque. Changing the gait to amble can increase the payload capacity by 180-200g, and further increases are possible with slower gaits like wave or ripple.

> 為確保足式機器人的安全與最佳效能，必須確定其最大酬載能力。Rex 所用的 Dynamixel 12A 馬達失速扭矩為 1.5Nm，我們針對承載酬載時最不利的三足步態進行計算。如圖 9 所示的分析顯示，即使在三足步態下，Rex 仍能安全承載其所有元件，相對於失速扭矩保有 25–33% 的安全餘裕。將步態改為緩行步態可使酬載能力增加 180–200 公克，若採用波動或漣波等更慢的步態，還能進一步提升。



(a) During tripod gait (b) During stance to walking

Fig. 9: Torque test results

(a) 三足步態期間 (b) 由站立轉為行走期間

圖 9：扭矩測試結果

3) Traversability Tests:

3) 通行能力測試：

To validate the performance in different and difficult terrains, Rex has been rigorously tested in various terrains as shown in Figure 10.

> 為驗證 Rex 在各種困難地形中的表現，我們在多種地形上對其進行了嚴格測試，如圖 10 所示。

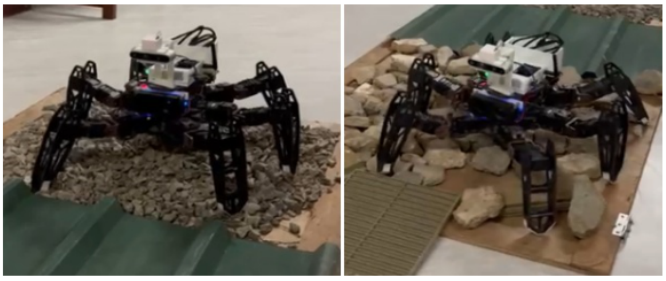


Fig. 10: Rex traversing in different terrains

圖 10：Rex 在不同地形中行進

V. Conclusion

五、結論

The proposed collaborative system involves an aerial and ground robot to explore and map environments, identify targets, deliver a medi-pack, and tele-conference with a victim. The system is low-cost, effective, and scalable to a team of robots, making it useful in search and rescue missions. The solution is easily adaptable to any off-the-shelf drone capable of carrying the drone add-on. The use of a six-legged robot provides benefits in stability and leg manipulation, and the camera mounted on it is useful for exploration tasks.

所提出的協同系統結合空中與地面機器人，用以探索並建構環境地圖、辨識目標、投送醫療包，以及與受困者進行遠端通話。本系統成本低、成效佳，且可擴展為機器人團隊，適用於搜救任務。此方案可輕易套用於任何能搭載此外掛模組的市售無人機。採用六足機器人帶來穩定性與腿部操作上的優勢，其上安裝的相機也有助於探索任務。

VI. Limitations and Future Work

六、限制與未來工作

The research aimed to improve odometry in low-light conditions by combining attitude estimation from an external IMU with visual odometry. However, this approach did not solve the problem of losing the primary odometry measurement source. To address this limitation, we plan to add another primary odometry measurement unit for improved mapping in darker and featureless environments. We also plan to upgrade the hexapod's motors and evaluate their methodology in larger outdoor environments. Moreover, the proposed solution will be expanded to research areas such as multi-robot mapping and exploration. Finally, we aim to develop DroneViz as a commercial low-cost and low-weight drone add-on.

本研究試圖藉由結合外部 IMU 的姿態估測與視覺里程計，來改善低光照條件下的里程計表現。然而，此方法並未解決主要里程計量測來源遺失的問題。為解決此限制，我們計畫加入另一個主要里程計量測單元，以改善在較暗與缺乏特徵環境中的建圖效果。我們也計畫升級六足機器人的馬達，並在更大的戶外環境中評估其方法。此外，所提方案將延伸至多機器人建圖與探索等研究領域。最後，我們的目標是將 DroneViz 發展為商用的低成本、輕量化無人機外掛模組。

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